AP/ITEC 4220 A - Modern Approaches to Data Management: Database Management Systems

Milestone 1

Group 2

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Problems:

As the number of gaming consoles and video titles continues to expand, small video game stores face growing challenges in accurately tracking sales and managing inventory. Many still rely on traditional practices, such as point-of-sale systems combined with manual stock counts, which often result in errors and inefficiencies. Implementing an online database system will make shopping easier for customers and help stores manage their inventory more effectively.

Scope:

The database will maintain and enforce data related to the product catalog, inventory, customers, orders, and suppliers. It will also integrate with an e-commerce platform to enable online purchasing, ensure accurate inventory tracking, streamline restocking processes, and provide reliable access to information for both customers and store staff.

Business Rules:

- **Customer:** Customers have a unique ID and must provide personal details such as name, contact, and email.
- **Account:** A customer must register an Account before using checkout, receipts, or order history.
- **Shopping Cart:** A ShoppingCart is managed by exactly one Account, but can exist independently as a guest session.
- **Order**: To make a purchase an Order must include at least one CartLine. Once an order is finalized, a receipt is generated. An Account can have multiple orders.
- **CartLine:** Each CartLine refers to a single product within the shopping cart, along with its quantity and price.
- **PaymentMethod**: An Account must have at least one valid PaymentMethod (Card or E-Transfer) to complete checkout.

- **Product:** Each Product has a unique id, name, and belongs to a category (Video Game, Console, Accessory), and price.
- **Category**: Each product belongs to a single category (Video game, Consoles, Accessories)
- **Inventory:** The system continuously monitors inventory levels, maintaining records of each product's minimum and maximum thresholds and notifies the supplier whenever stock falls below the required level.
- **Suppliers:** Suppliers have a SupplierContract to provide games from their catalog and are responsible for shipping when inventory falls below the defined minimum. Suppliers information, including contact, and pricing is stored in the system.

User Views:

Customers

Data requirements

- Customer ID, name, contact, email, address
- Payment method (card, e-transfer)
- Orders (purchase)

Information requirements

- Browse games by name, category, availability, and price
- Track order status (pending, completed)
- Access receipts and past purchase history

Suppliers

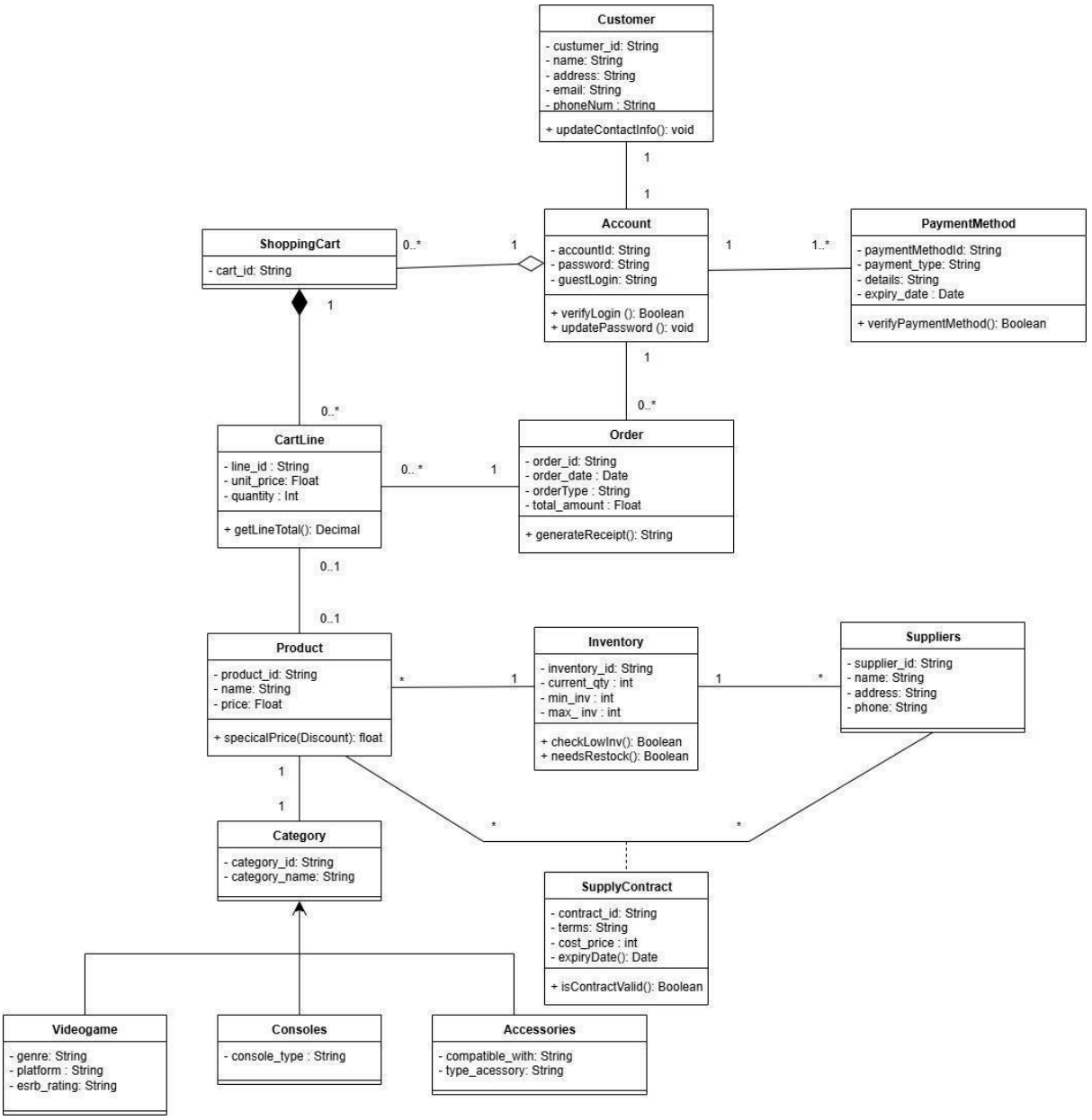
Data requirements

- Supplier ID, Name, Contact details
- Product catalog (games offered, pricing, categories)
- Store restock requests (game, quantities, deadlines)
- Shipping records (deliveries to store)

Information requirements

- Manage incoming stock requests from stores
- Track shipment to stores
- Supplier Contract
- Update game availability and pricing in catalog

Class Diagram



Video Game store System

ITEC 4220

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Problems

Traditional sales and inventory tracking methods are becoming increasingly ineffective. As the number of consoles and video games grows, stores struggle to maintain accurate records and efficient restocking, leading to various operational challenges.



High Error Rates	Rising consoles and games complicate inventory tracking	Manual processes lead to frequent mistakes	Stockout and overstocking of popular items
Inefficient Inventory Management	Challenges in restocking items in a timely manner	Limited visibility across different store locations	Ineffective use of space and resources

Business Scope

This system manages all aspects of the store, promoting efficiency and organization. Its two core components, Inventory Management and Order Processing, simplify operations and improve the customer experience



Inventory Management	Track real-time stock levels effectively	Maintain a comprehensive product catalog	Centralized database for quick product updates
Order Processing	Maintain customer accounts and purchase details	Apply discounts and handle multiple payment methods easily	Streamline supplier orders through E-commerce integration

Business Rules

- **Customer:**
  - Unique ID.
  - Personal details: name, contact, and email
- **Account:**
  - Customer must have an account & valid payment method
  - Customer can also have guest login.
- **Shopping Cart:**
  - Can be linked to exactly one account
  - Zero or more products
  - Can add items without an account using guest login
- **Order:**
  - Represent an purchase
  - Contain at least one or more CartLines
  - Generated a receipt
- **Cartline**
  - Refers to a line in shopping cart (quantity & price)

Business Rules:

- **PaymentMethod:**
  - At least one of it
  - Accept: E-transfer, card ONLY
- **Product:**
  - UID
  - Name, category, price
- **Category:**
  - Each product belongs to “SINGLE” category
  - E.g. [Video game], [Consoles], [Accessories]
- **Inventory:**
  - Multiple products are stored in one inventory
  - Maintain record (MAX & MIN) thresholds
  - Falls below required level -> initiates supplier restock request
- **Supplier:**
  - Multiple Suppliers supply multiple products to the inventory
  - Has a Supplier Contract for terms with which product to ship

User Views (Customers)

Data requirements

- Customer ID, name, contact, email, address
- Payment method (card, e-transfer)
- Orders (purchase)

Information requirements

- Browse games by name, category, availability, and price
- Track order status (pending, completed)
- Access receipts and past purchase history

User Views (Suppliers)

Data requirements

- Supplier ID, Name, Contact details
- Product catalog (games offered, pricing, categories)
- Store restock requests (game, quantities, deadlines)
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Information requirements

- Manage incoming stock requests from stores
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Class Diagram:

